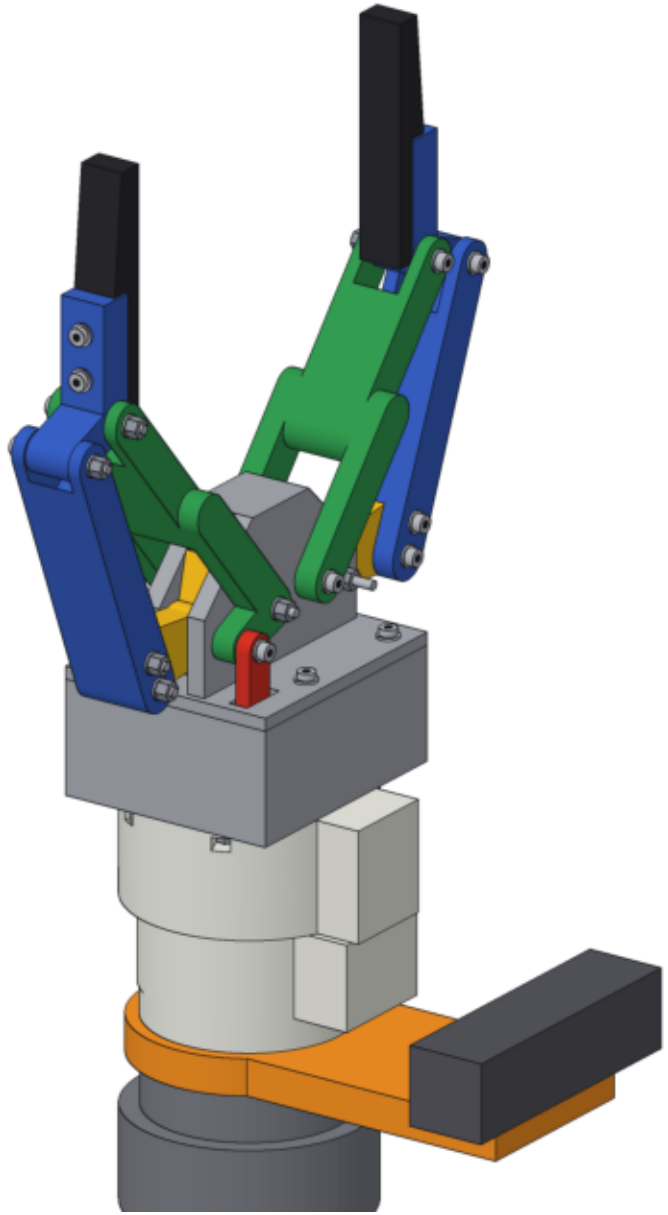
HomeMade2F140

Manual de ensamble



Gripper de 2 dedos con la cinematica del Robotiq 2F-140

Servo DFRobot SER0062 (D5545GS) - version v2

17 piezas impresas - 36 tornillos M3 - 4 tornillos M6

30 tuercas M3 nyloc - 50 arandelas M3

Herramientas: llave allen 2,5 mm y 5 mm,

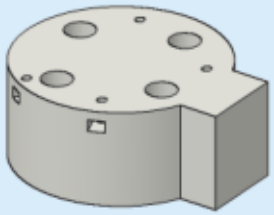
llave de 5,5 mm para tuercas M3.

Pasos 1-12: muneca y caja con el servo

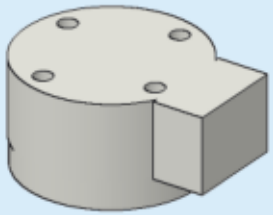
Pasos 13-23: torre con los dedos (en la mesa)

Pasos 24-26: union final

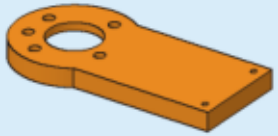
Inventario de piezas



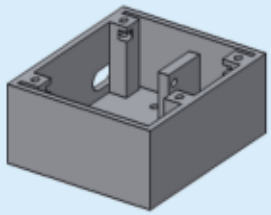
1x Espaciador B



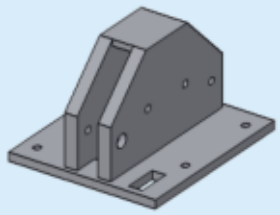
1x Espaciador A



1x Camera mount



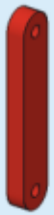
1x Caja



1x Torre



1x Manivela



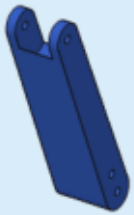
1x Biela



1x OuterKnuckle_L



1x OuterKnuckle_R



2x OuterFinger



2x InnerKnuckle



2x InnerFinger



2x Fingertip

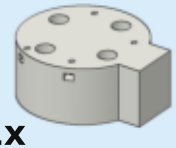
Tornilleria

6x M3 x 8	2x M3 x 45
9x M3 x 12	4x M6 x 75
9x M3 x 16	30x tuerca M3 nyloc
10x M3 x 30	50x arandela M3 (7 x 0,5 mm)

STL: carpeta print/parts del repositorio (la cantidad esta en el nombre del archivo, _x2 = 2 piezas).

1

Tuercas en el espaciador B

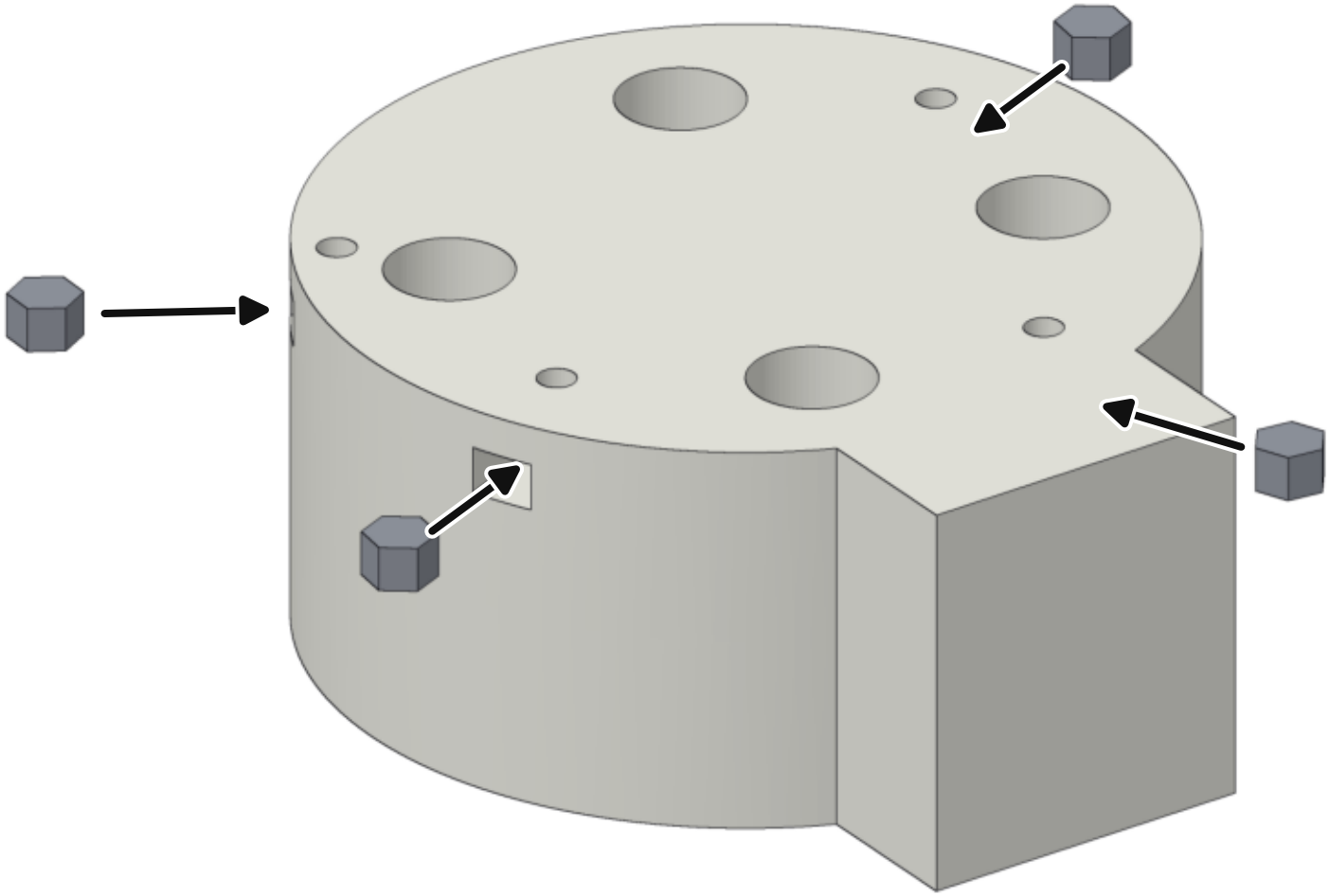


1x
Espaciador B



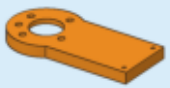
4x
Tuerca M3 nyloc

- Mete una tuerca nyloc en cada ranura lateral hasta que quede centrada con el agujero.



2

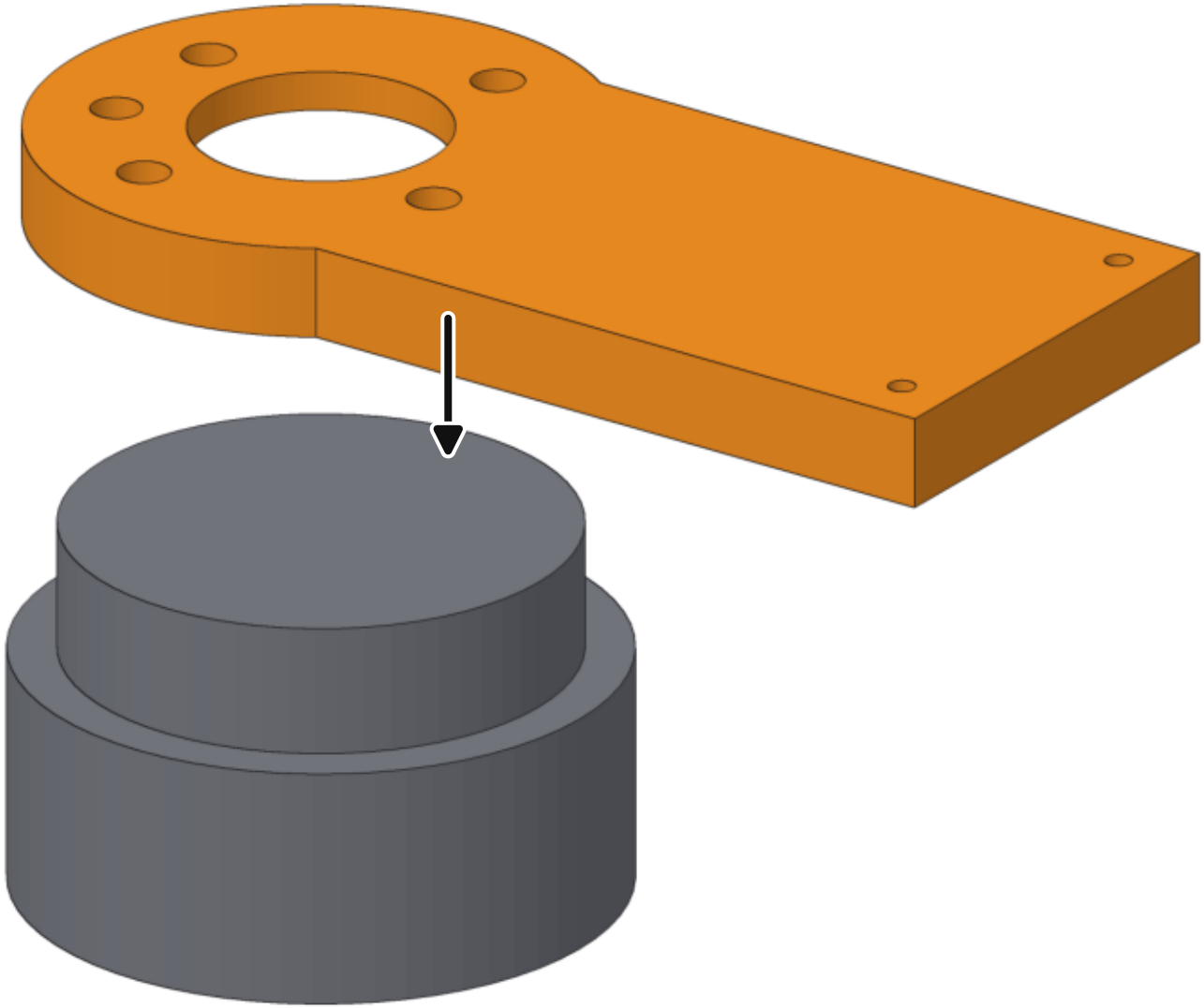
Camera mount sobre la brida del robot



1x
Camera mount

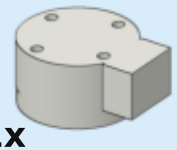
- El brazo de la camara apunta hacia +X (el lado donde ira la biela roja).

- Opcional: pasador de 6 mm en el agujero de centrado.

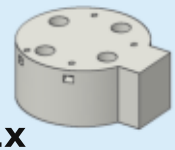


3

Espaciadores A y B

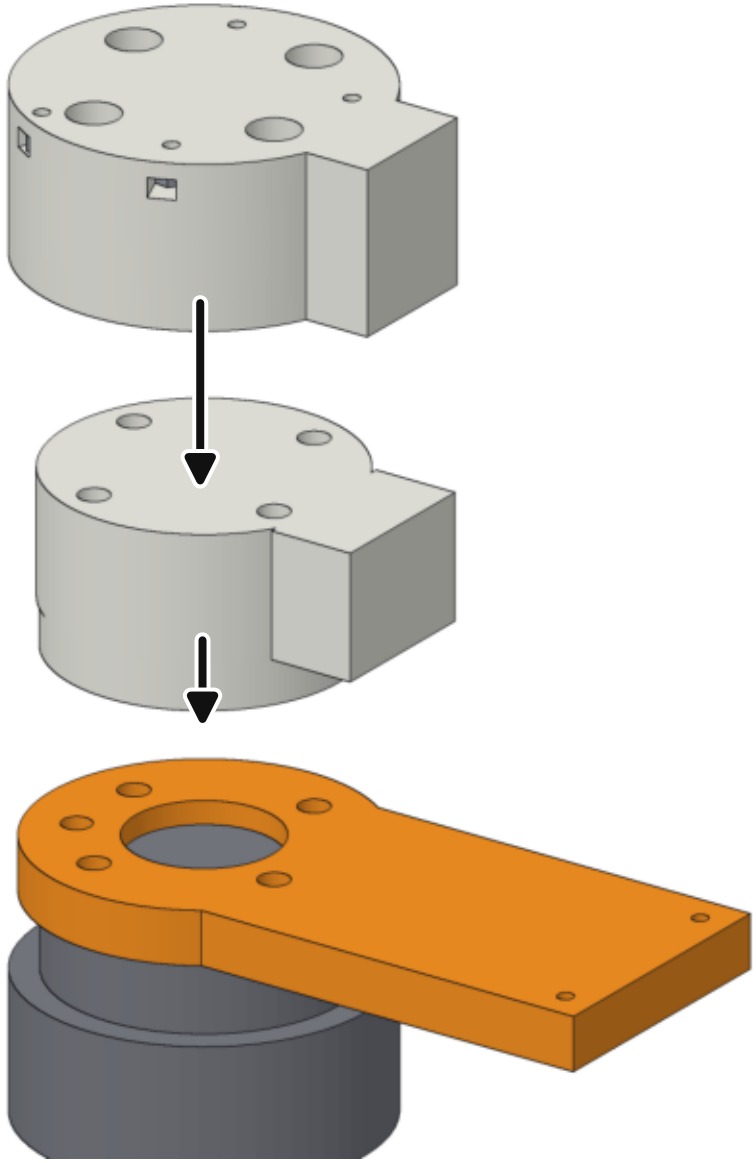


1x
Espaciador A



1x
Espaciador B (paso 1)

- El bloque saliente de ambos espaciadores va hacia el brazo de la camara.



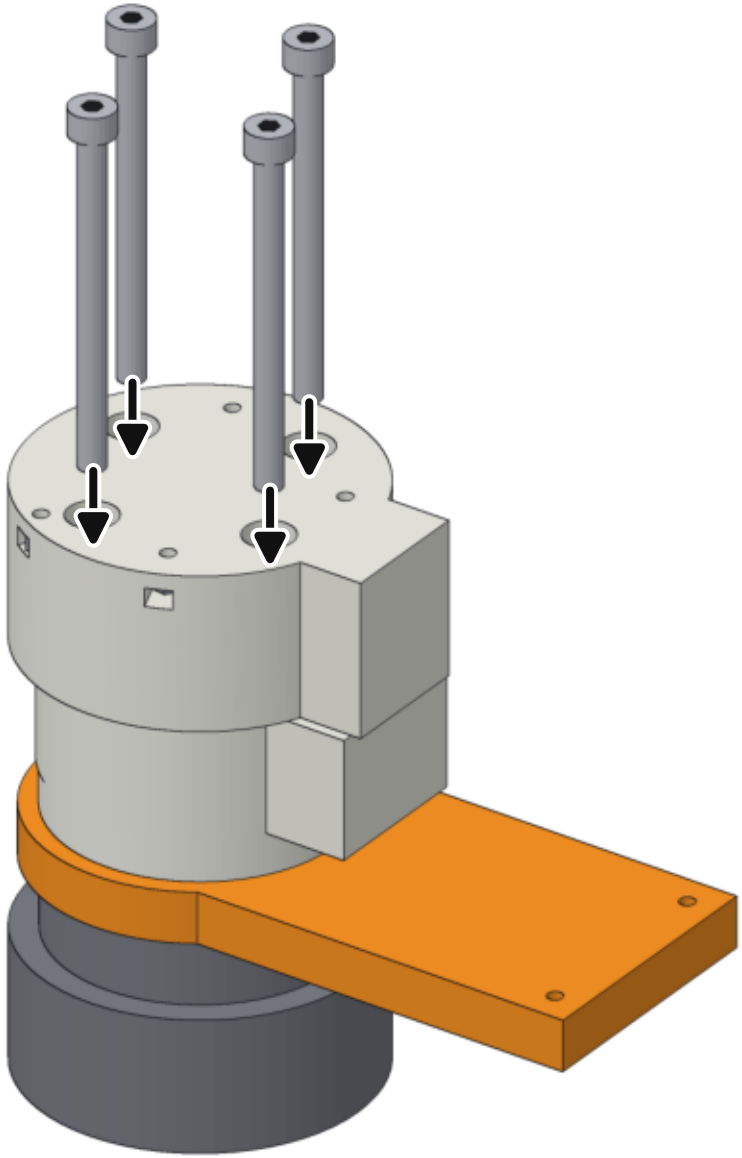
4

4 tornillos M6 a la brida



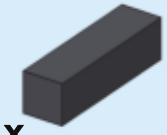
4x
M6 x 75

- Aprieta en cruz. La cabeza queda hundida en el espaciador B.

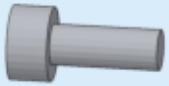


5

Camara RealSense D435i

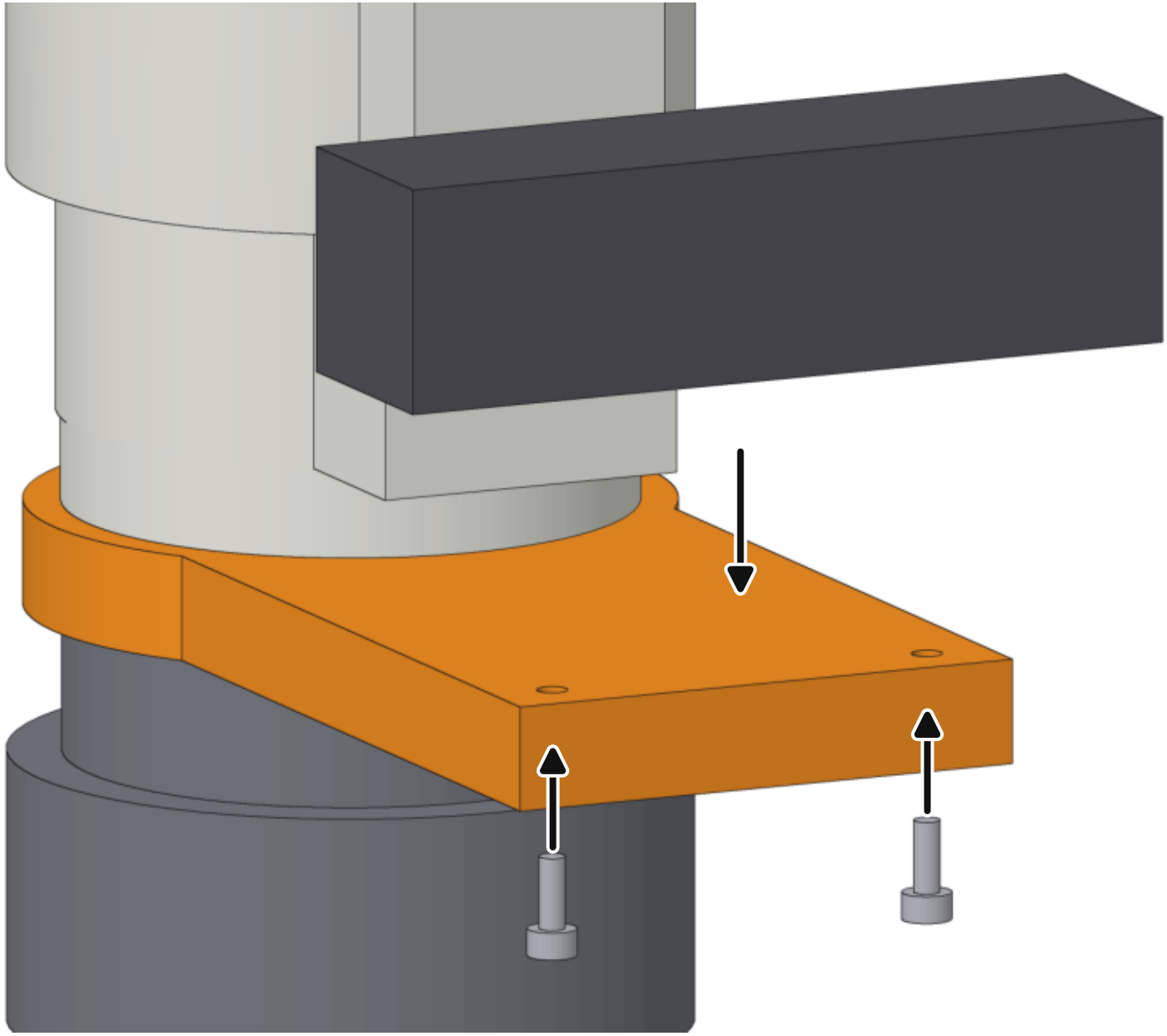


1x
RealSense D435i



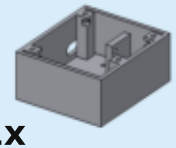
2x
M3 x 8

- Los 2 tornillos entran por debajo del camera mount.



6

Tuercas en la caja

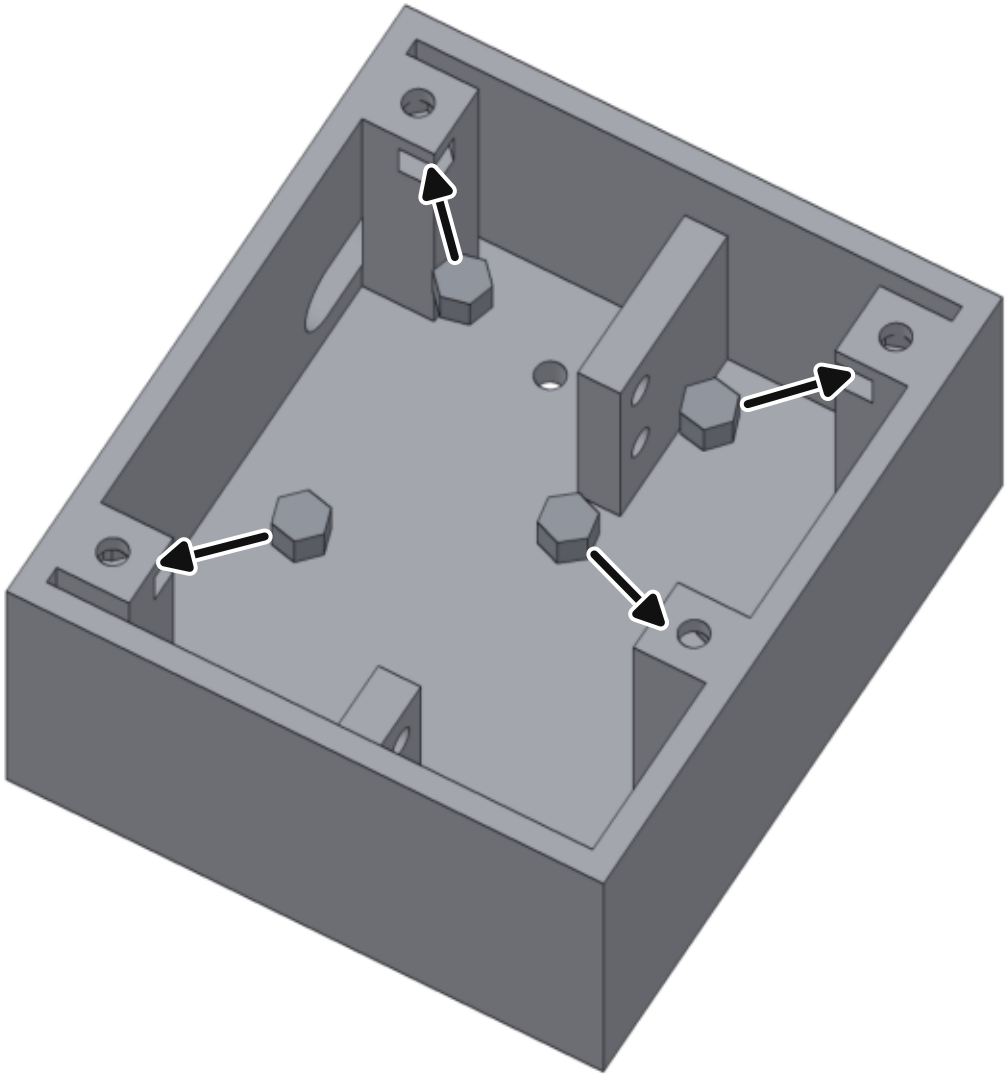


1x
Caja



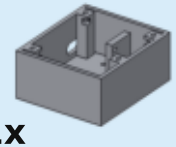
4x
Tuerca M3 nyloc

- Una tuerca nyloc en cada columna de las esquinas, por la ranura interior.



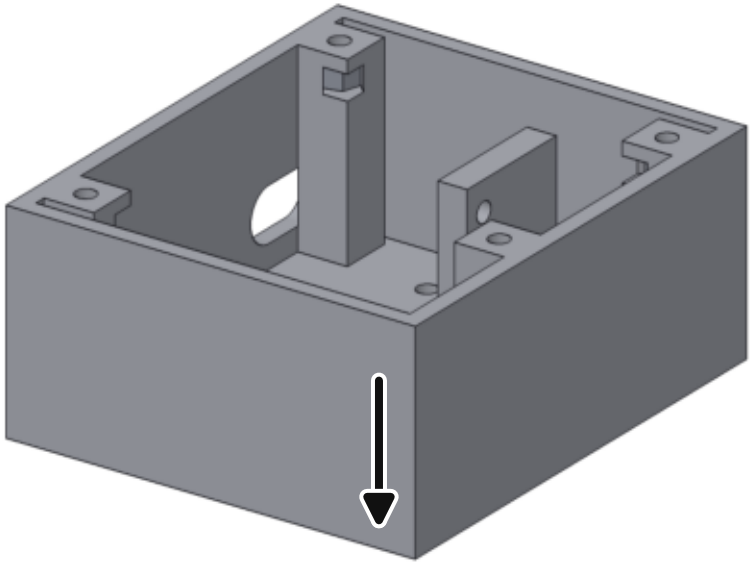
7

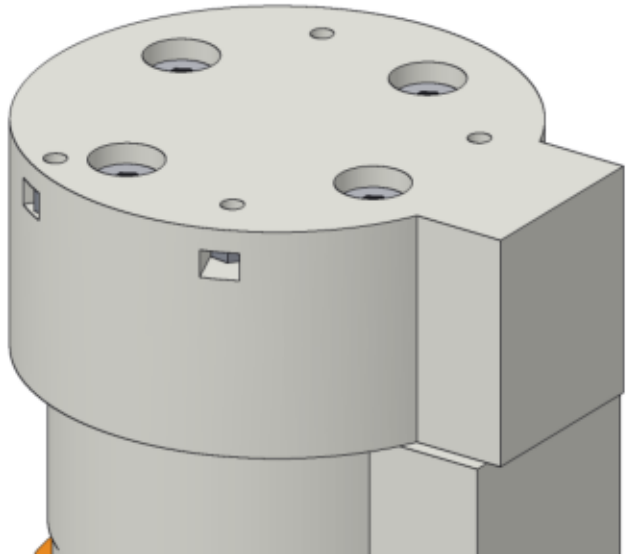
Caja sobre el espaciador B



1x
Caja (paso 6)

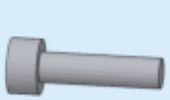
- La pared con el hueco de cable (-X) va del lado opuesto a la camara.



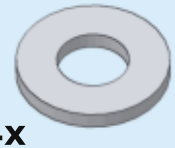


8

Tornillos de la caja

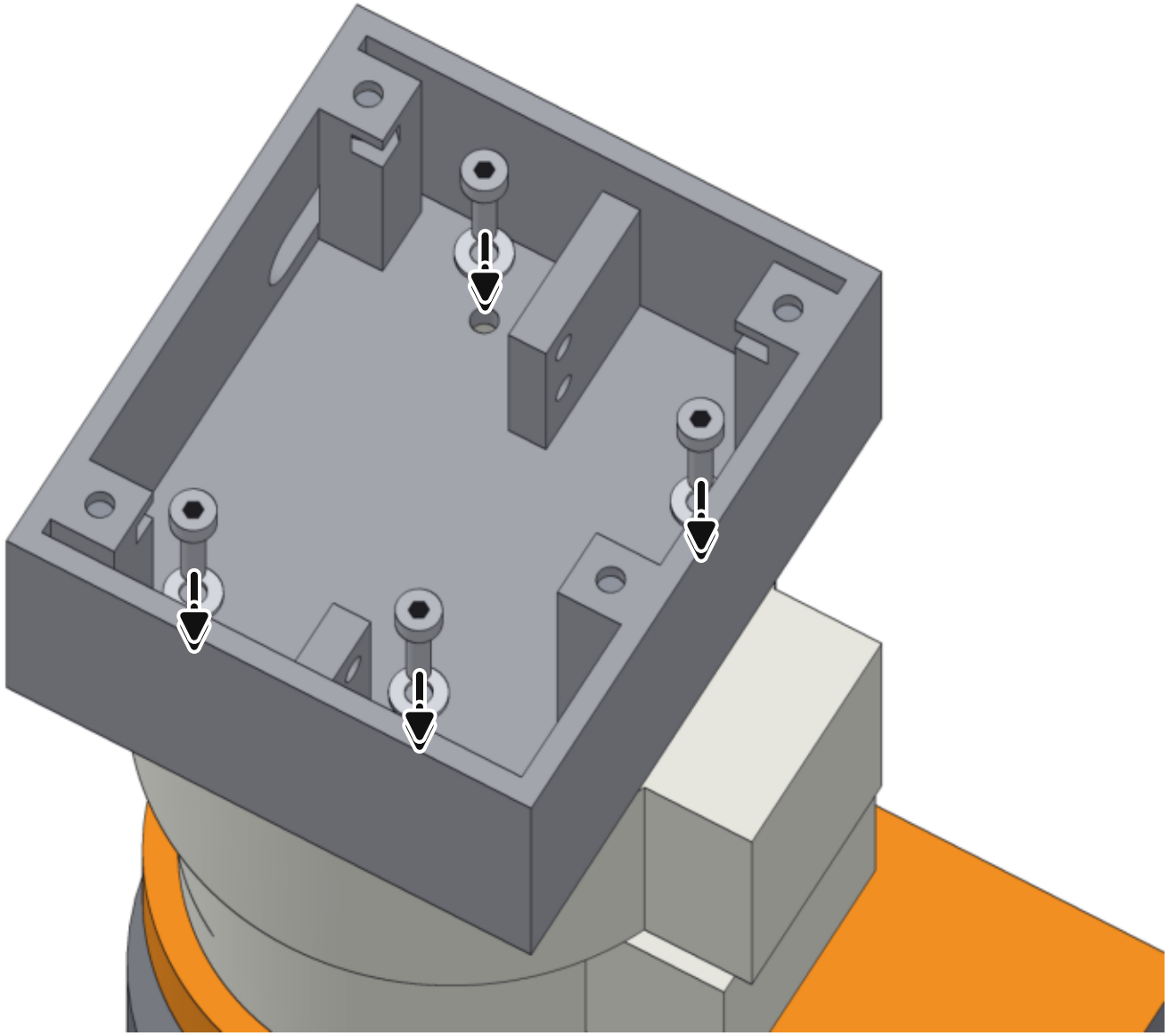


4x
M3 x 12



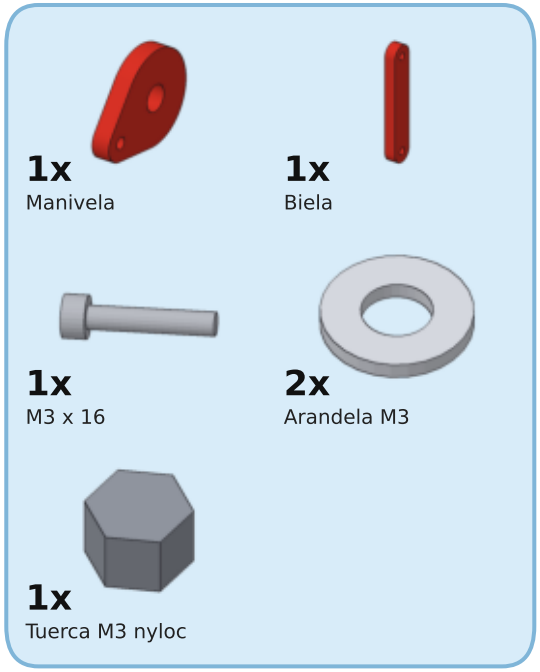
4x
Arandela M3

- Desde dentro de la caja, antes de meter el servo.

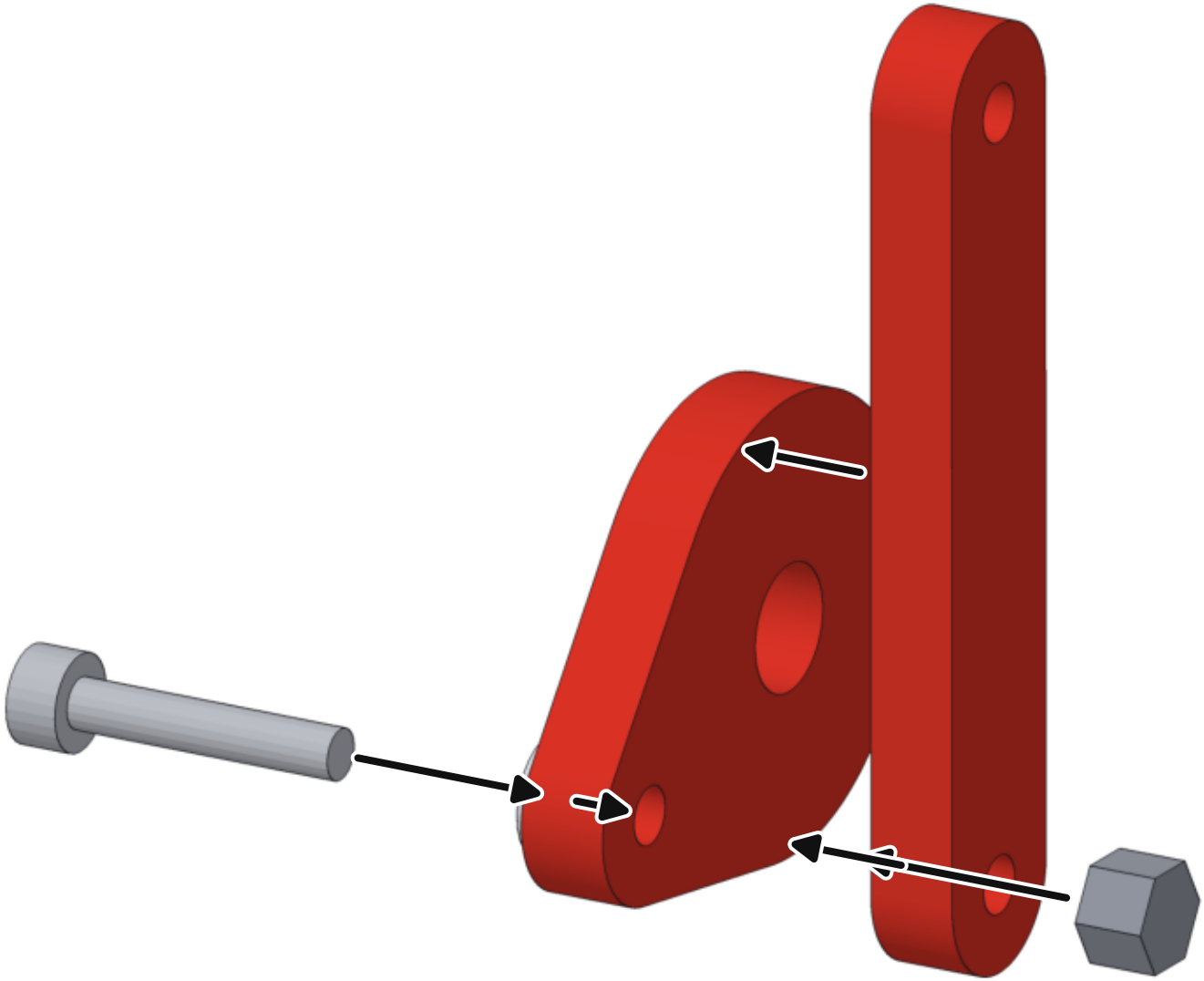


9

Manivela + biela (en la mesa)



- Aprieta la tuerca solo hasta quitar el juego: la biela debe girar libre.



10

Horn y manivela al servo



1x
Servo D5545GS



1x
Horn metalico

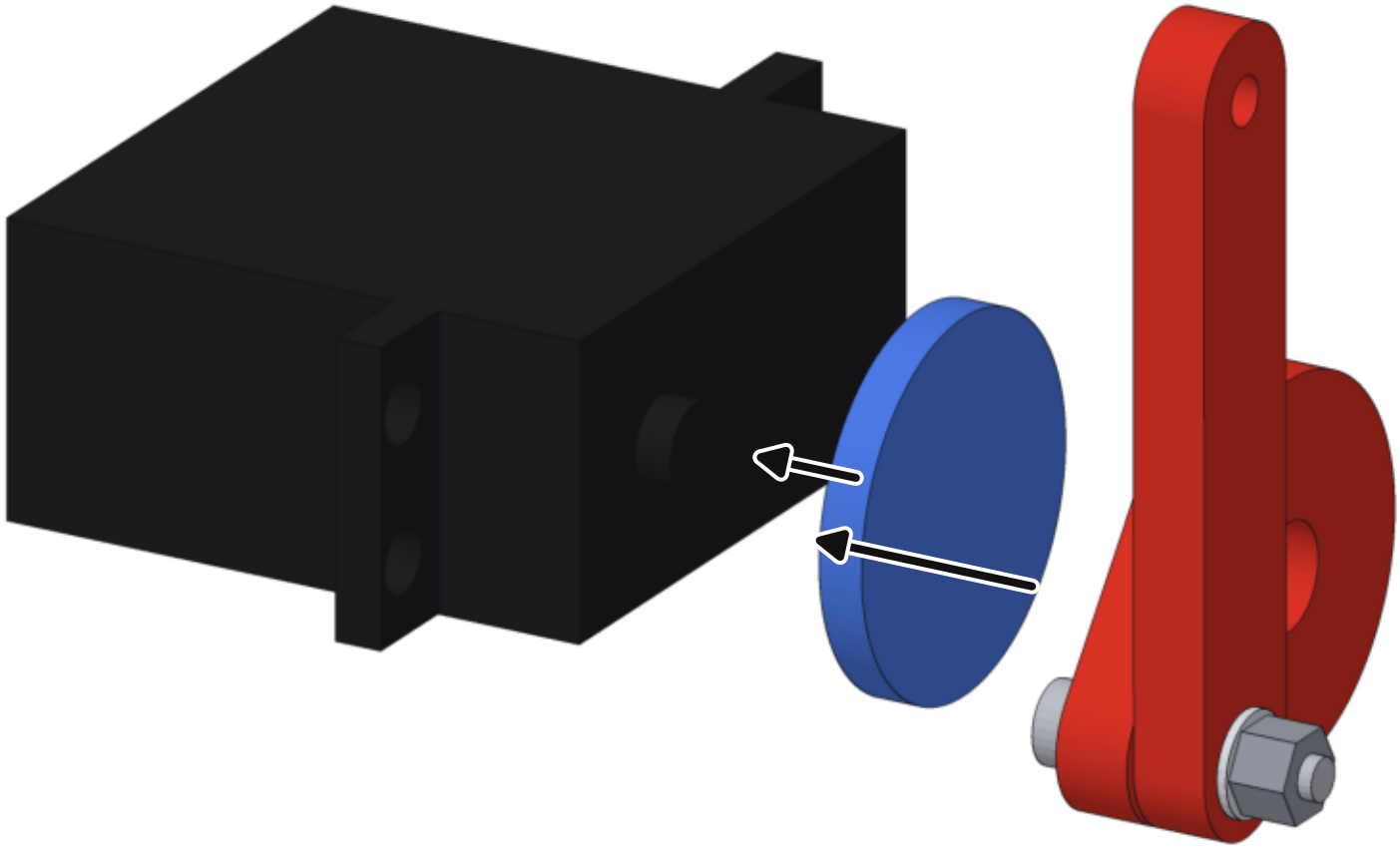


1x
Manivela + biela (paso 9)

- 1) Taladra la manivela usando el horn metalico como plantilla.

- 2) Lleva el servo a su centro (1500 us).

- 3) Monta el horn con la manivela hacia -Y (como en el dibujo) y la biela vertical.



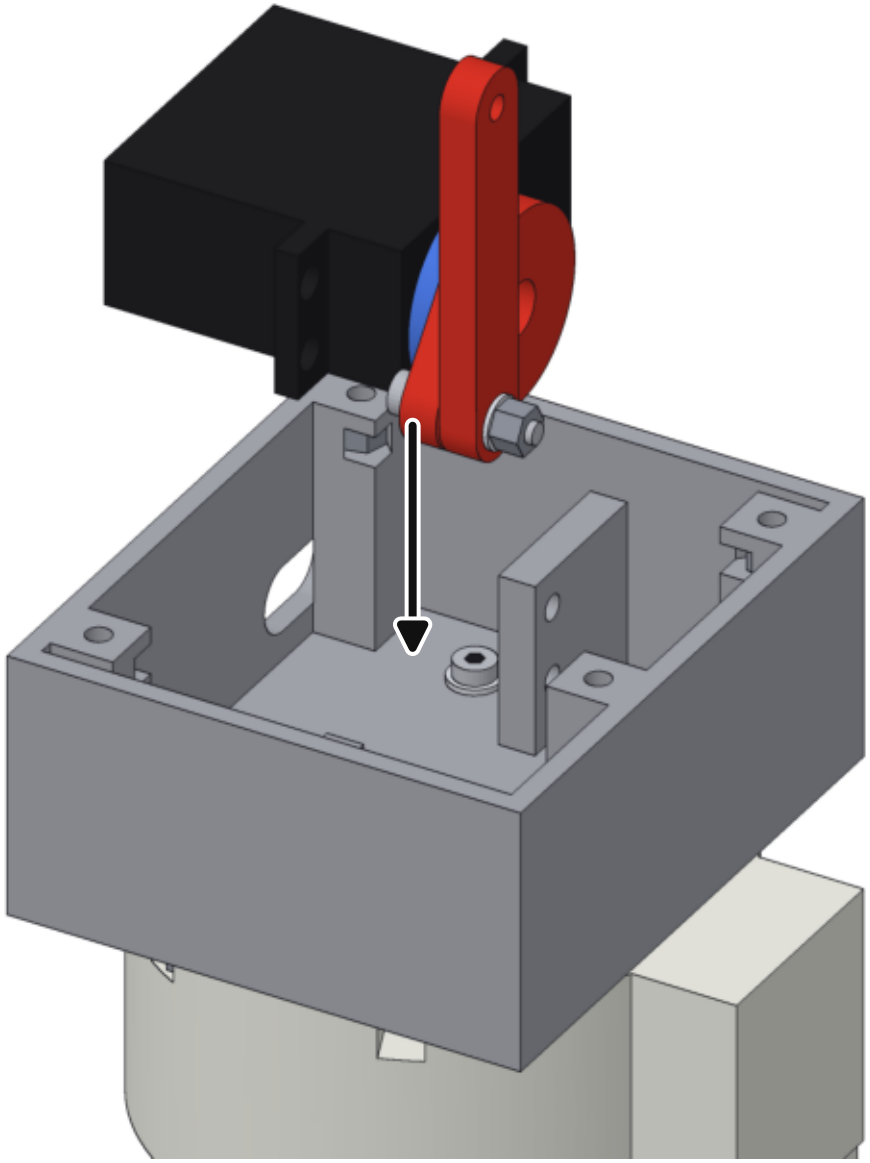
11

Servo dentro de la caja



1x
Servo + manivela (paso 10)

- Baja el servo desde arriba: las orejas apoyan en la mampara interior.

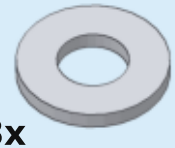


12

Tornillos del servo



4x
M3 x 16

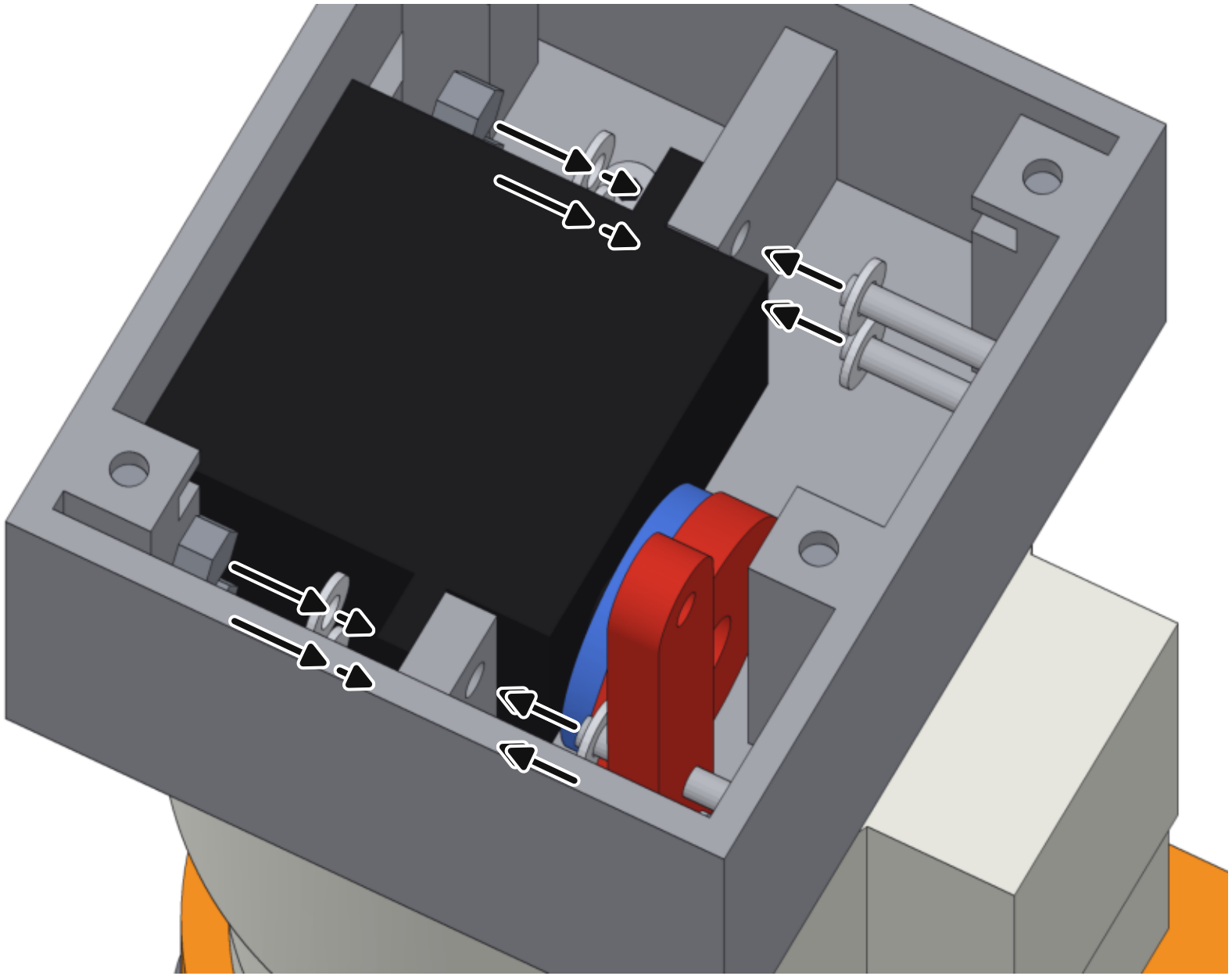


8x
Arandela M3



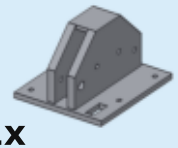
4x
Tuerca M3 nyloc

- Cabeza del lado de la mampara (+X), tuerca del lado de la oreja del servo.



13

Outer knuckles en la torre (en la mesa)



1x
Torre



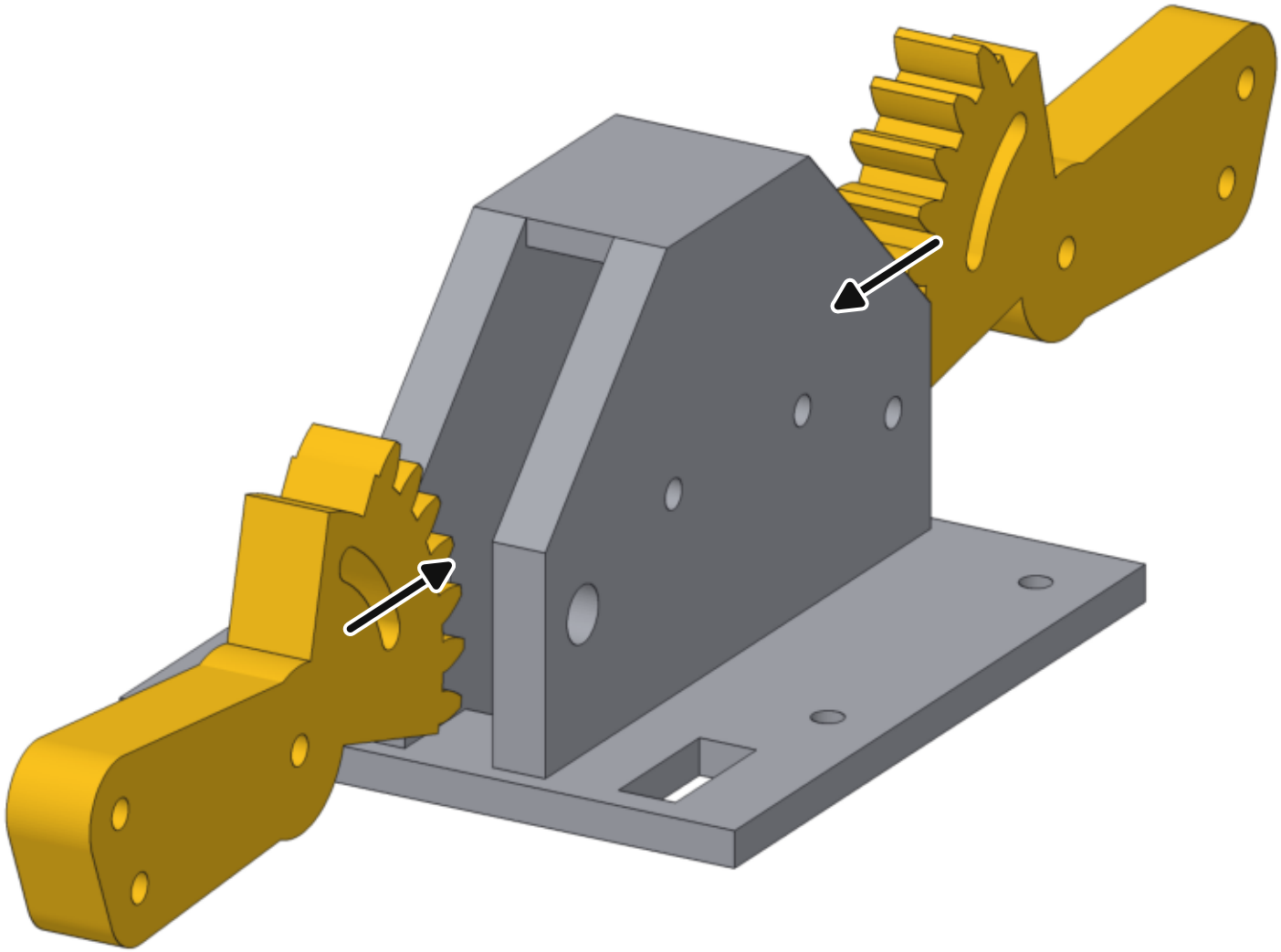
1x
OuterKnuckle_L



1x
OuterKnuckle_R

- OuterKnuckle_L va del lado -Y (lado de la biela). L y R NO son intercambiables.

- Engrana ambos con los dedos en posicion abierta (simetricos).

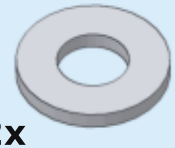


14

Pivotes de los outer knuckles



2x
M3 x 30

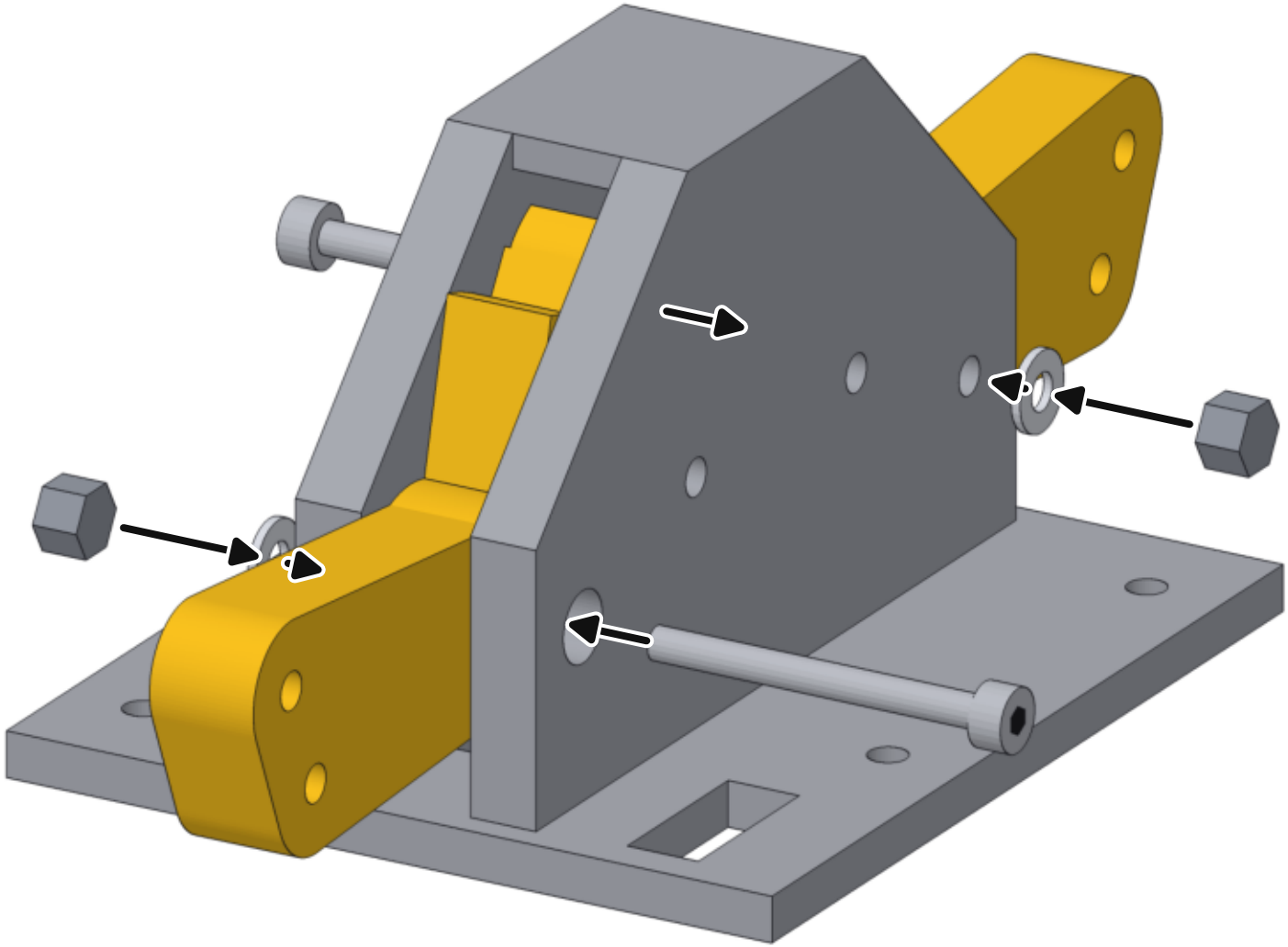


2x
Arandela M3



2x
Tuerca M3 nyloc

- Izquierdo: cabeza hundida en la cara +X.
  Derecho: cabeza hundida en la cara -X.



15

Tuerca en la oreja (en la mesa)

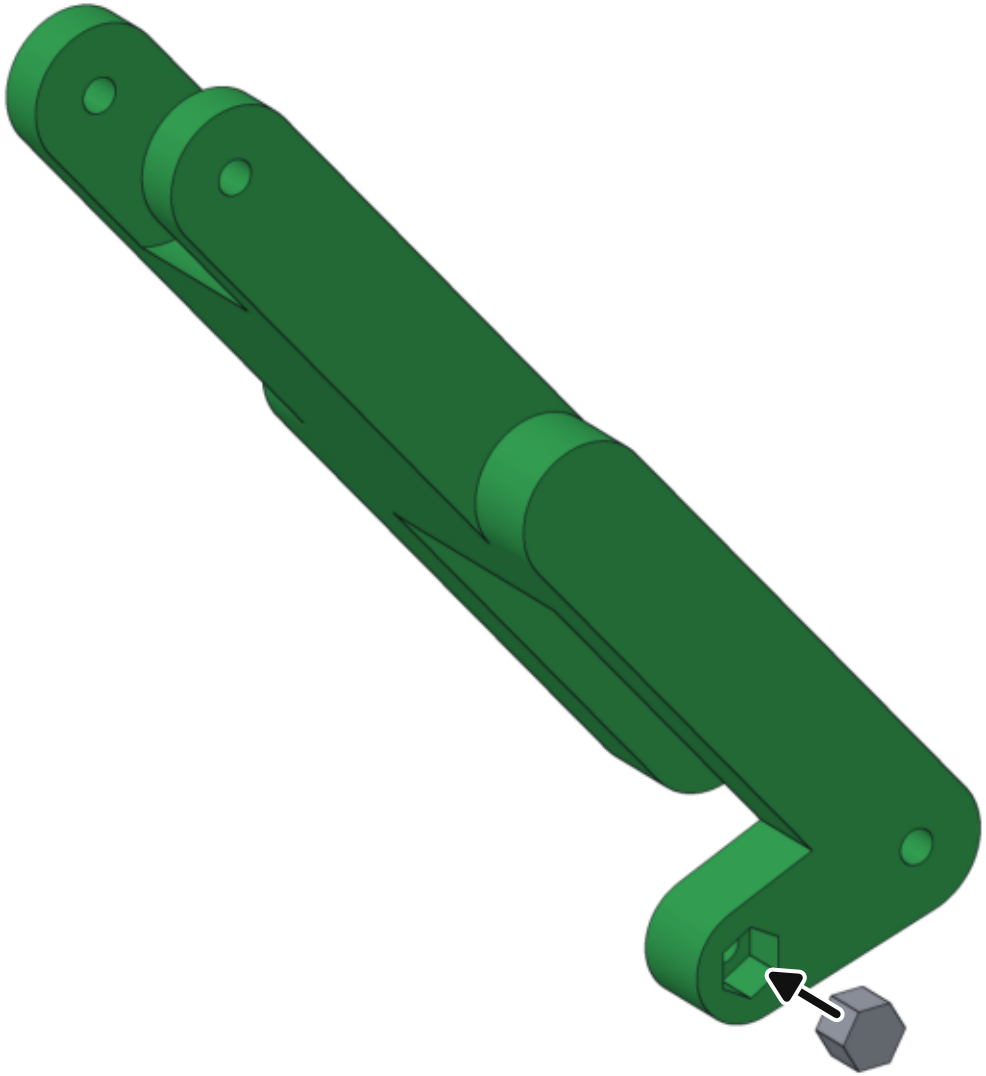


1x
InnerKnuckle



1x
Tuerca M3 nyloc

- Solo el inner knuckle que va del lado -Y usa la oreja. Mete la tuerca hasta el fondo del hexagono.



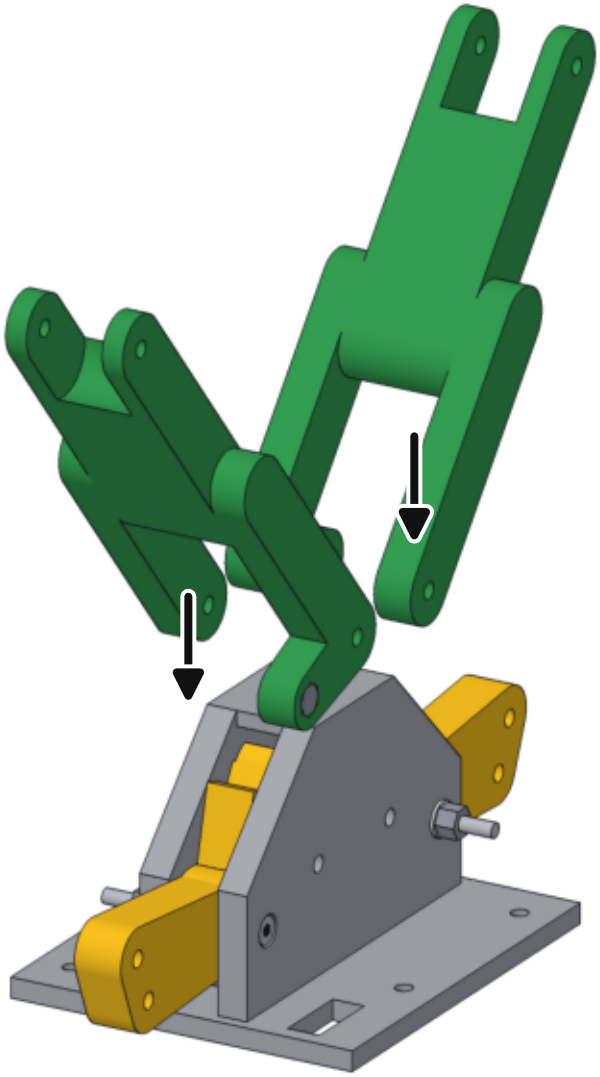
16

Inner knuckles



2x
InnerKnuckle

- Los brazos abrazan la torre por fuera. La oreja con tuerca queda del lado -Y, cara +X.

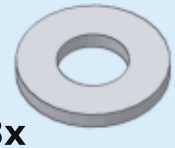


17

Pivotes de los inner knuckles



2x
M3 x 45

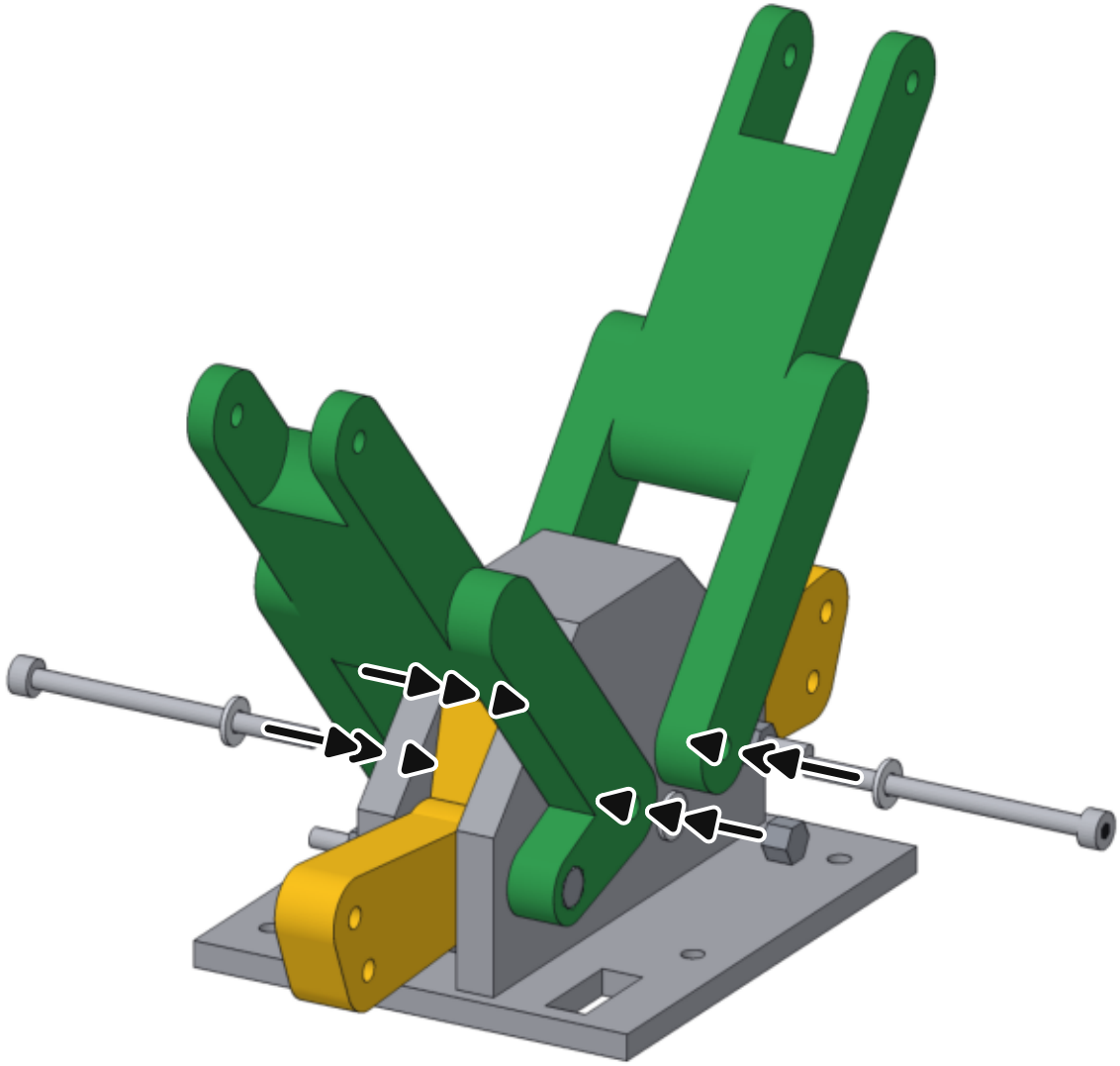


8x
Arandela M3



2x
Tuerca M3 nyloc

- Orden: brazo - arandela - pared - ranura del engrane - pared - arandela - brazo.



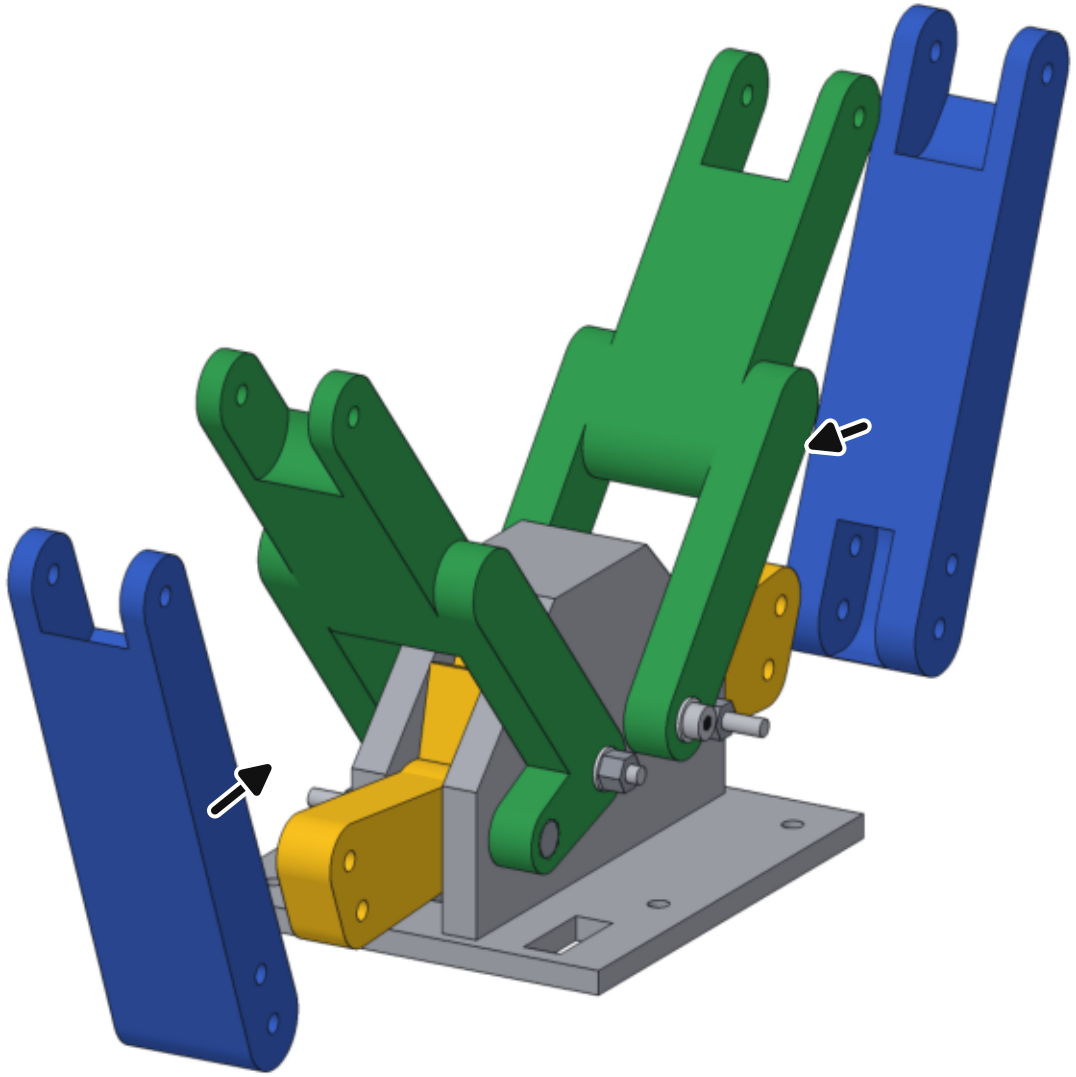
18

Outer fingers



2x
OuterFinger

- La horquilla del outer finger abraza el brazo amarillo.

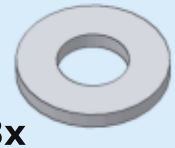


19

Union outer knuckle - outer finger



4x
M3 x 30

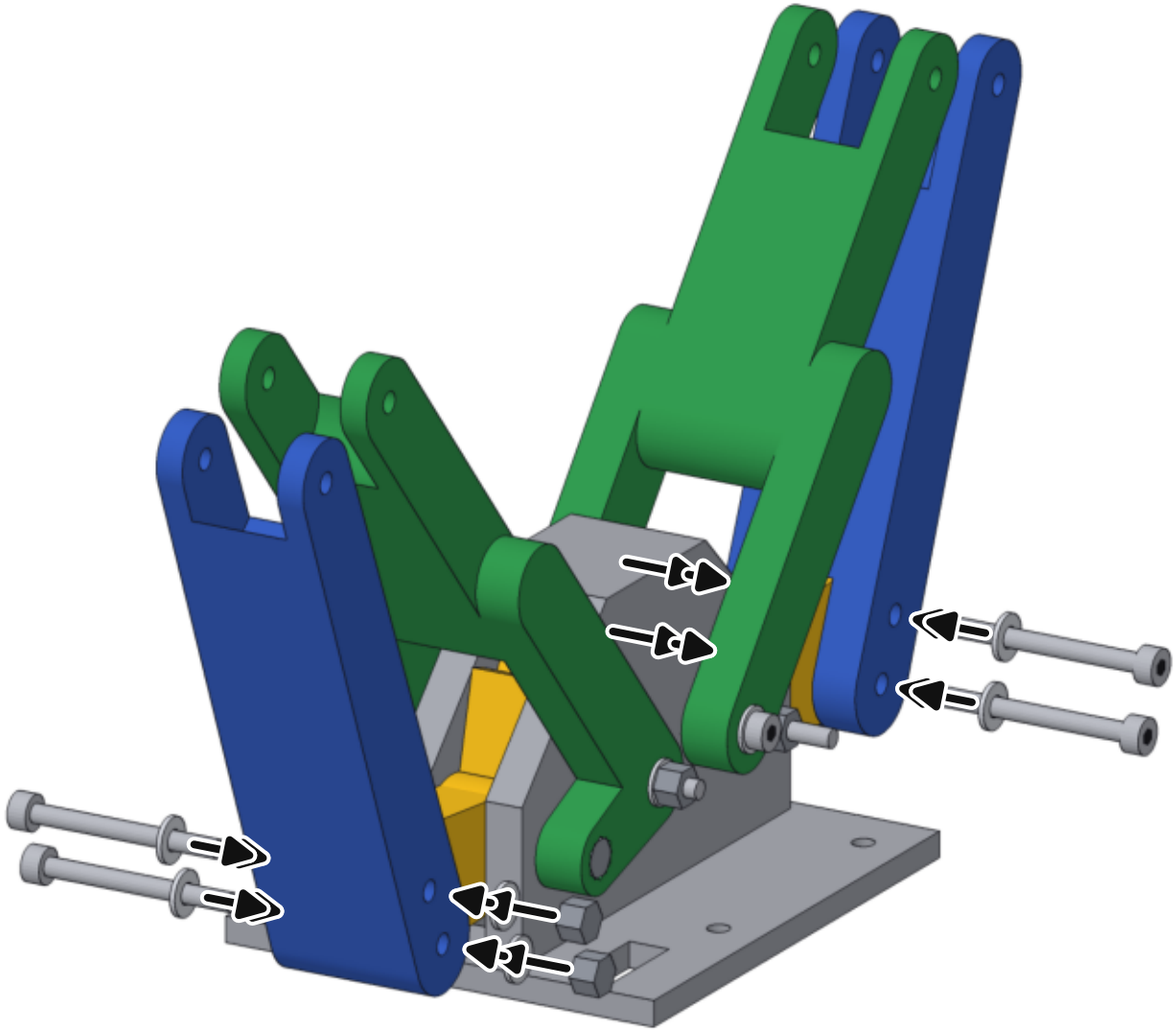


8x
Arandela M3



4x
Tuerca M3 nyloc

- 2 tornillos por lado: esta union es rigida, aprieta firme.



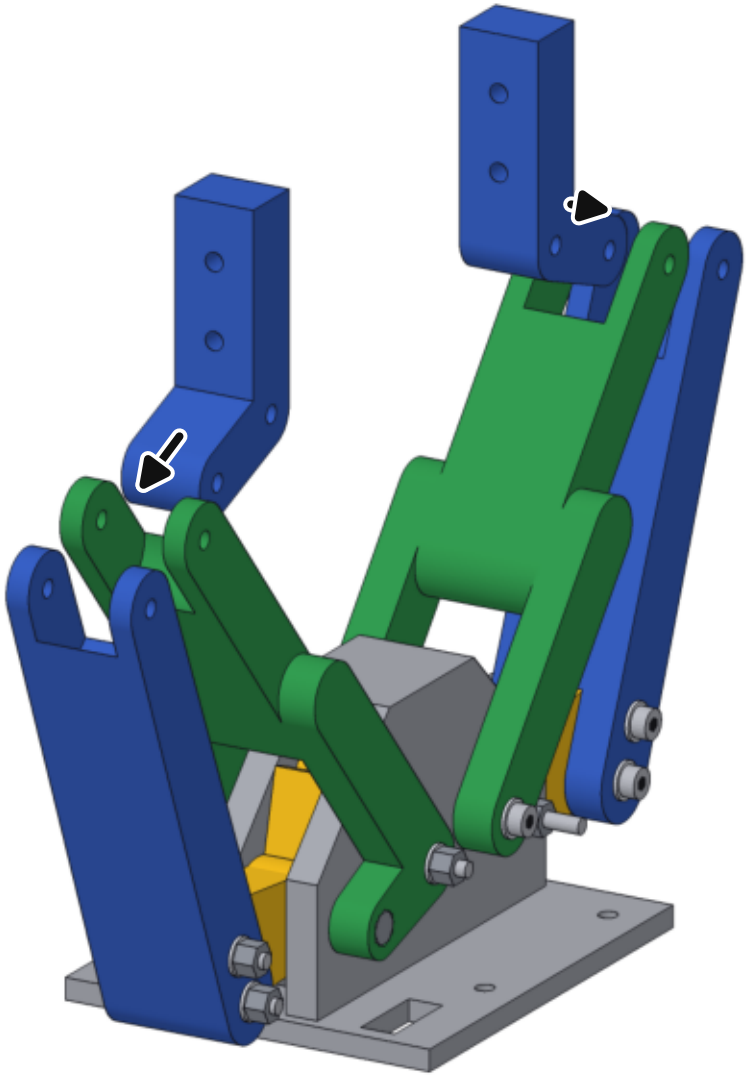
20

Inner fingers



2x
InnerFinger

- Entran en las horquillas del outer finger (azul) y del inner knuckle (verde).

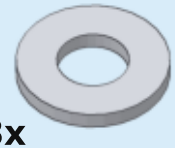


21

Pivotes de los inner fingers



4x
M3 x 30

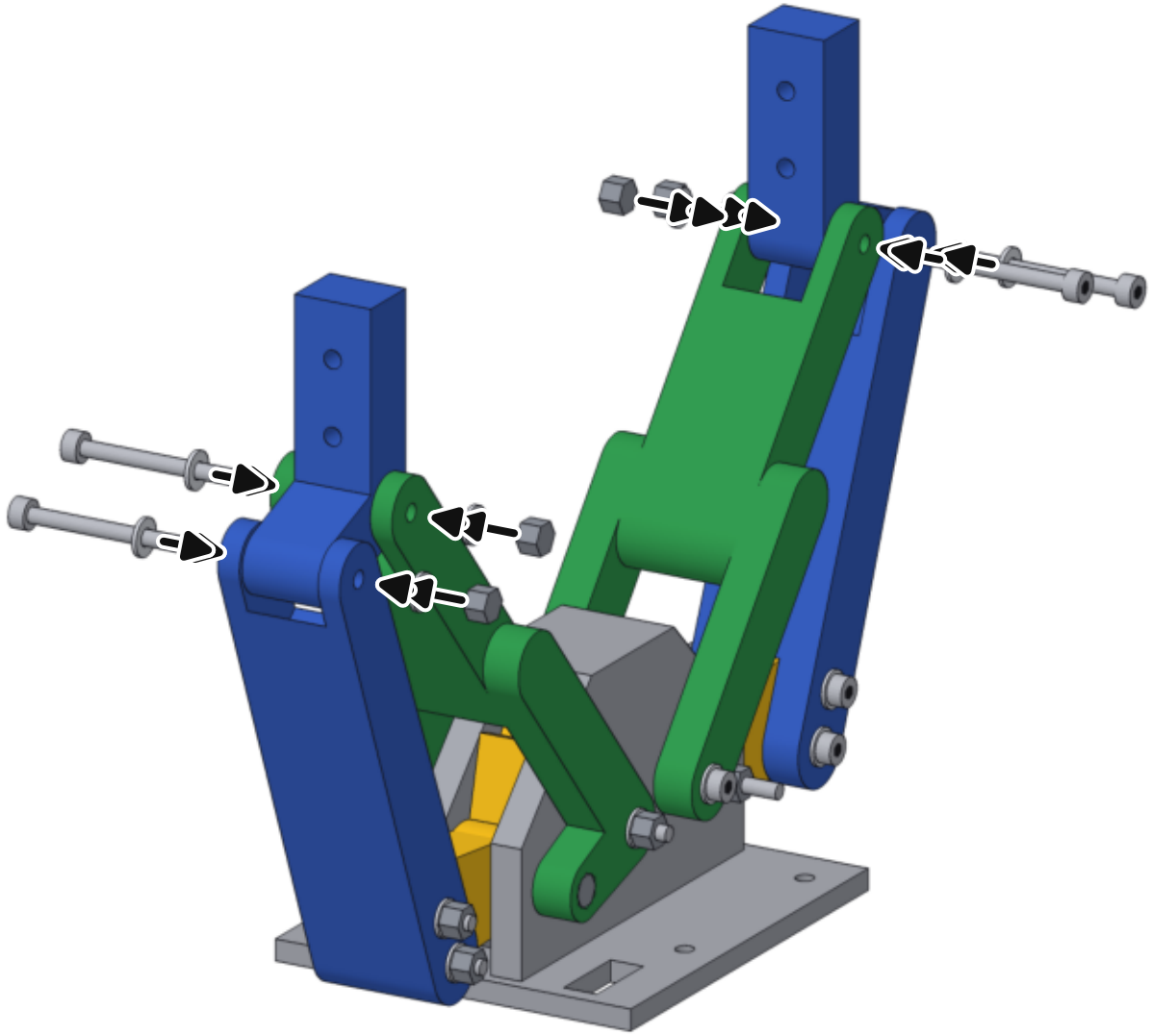


8x
Arandela M3



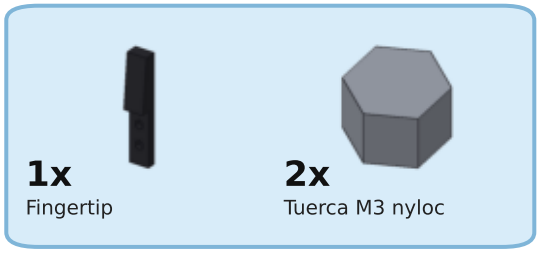
4x
Tuerca M3 nyloc

- Aprieta hasta quitar el juego; los dedos deben moverse suaves.



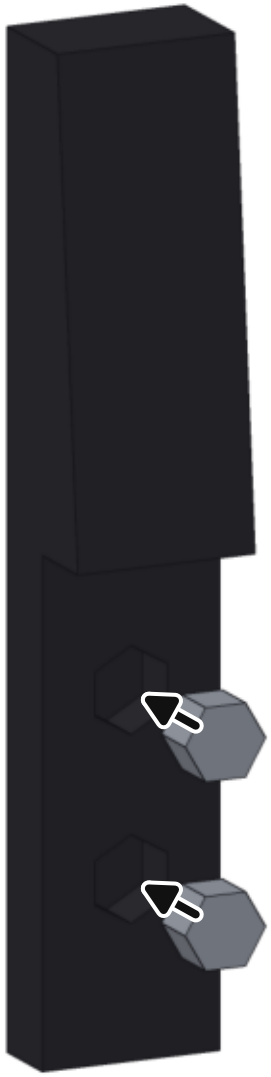
22

Tuercas en las puntas (en la mesa)



- 2 tuercas en los hexagonos de la cara trasera de cada punta.

- Repite con la otra punta.



x2

23

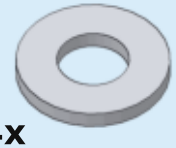
Puntas en los inner fingers



2x
Fingertip

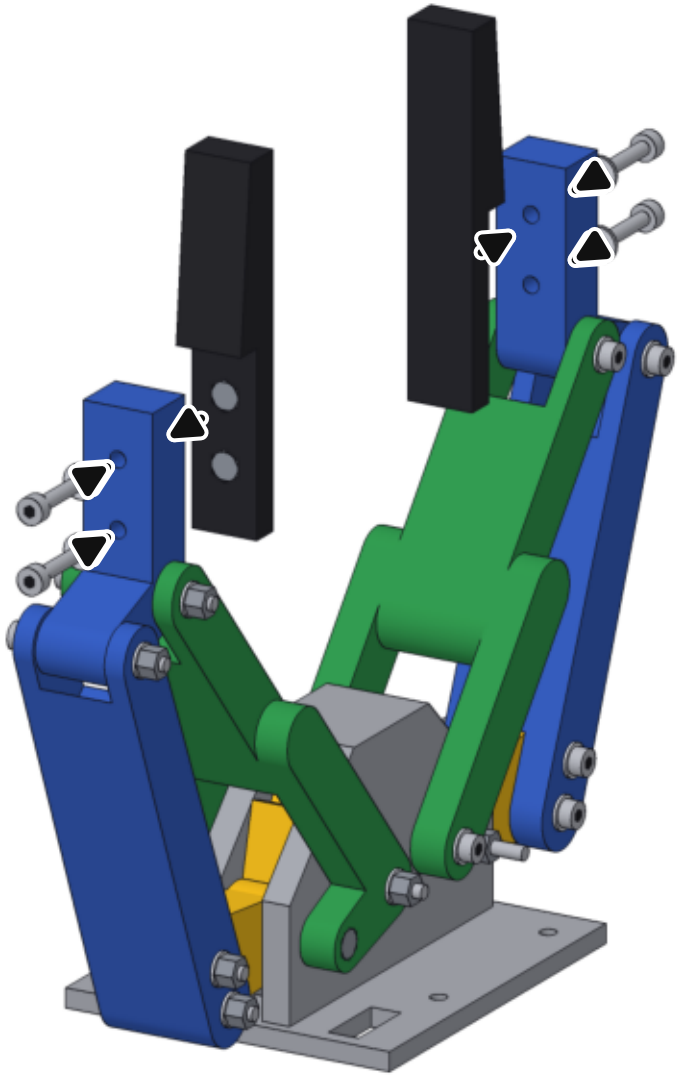


4x
M3 x 16



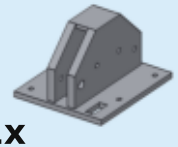
4x
Arandela M3

- Tornillos desde la cara exterior del inner finger.



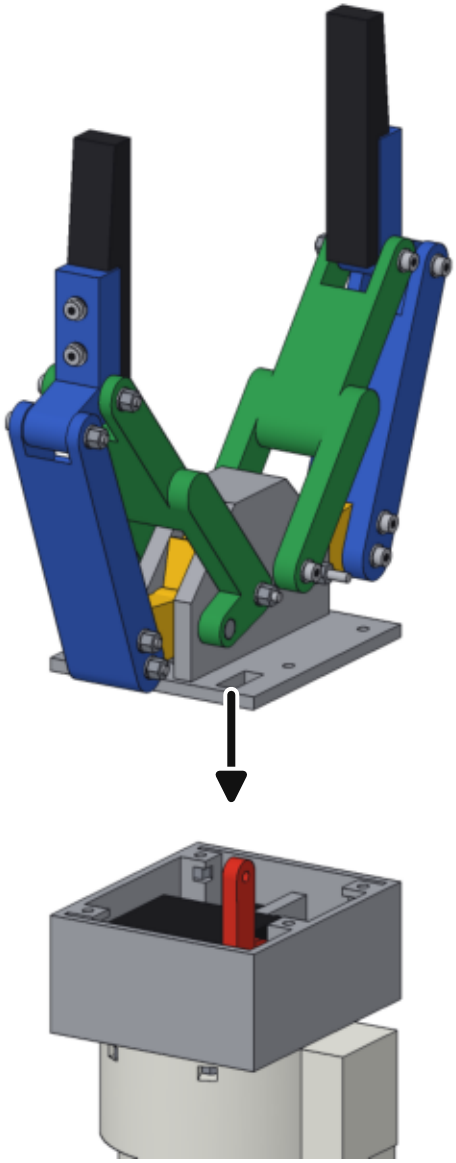
24

Torre sobre la caja



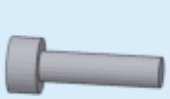
1x
Torre con dedos (pasos 13-23)

- Pasa la biela roja por la ranura de la placa de la torre.

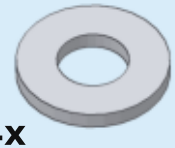


25

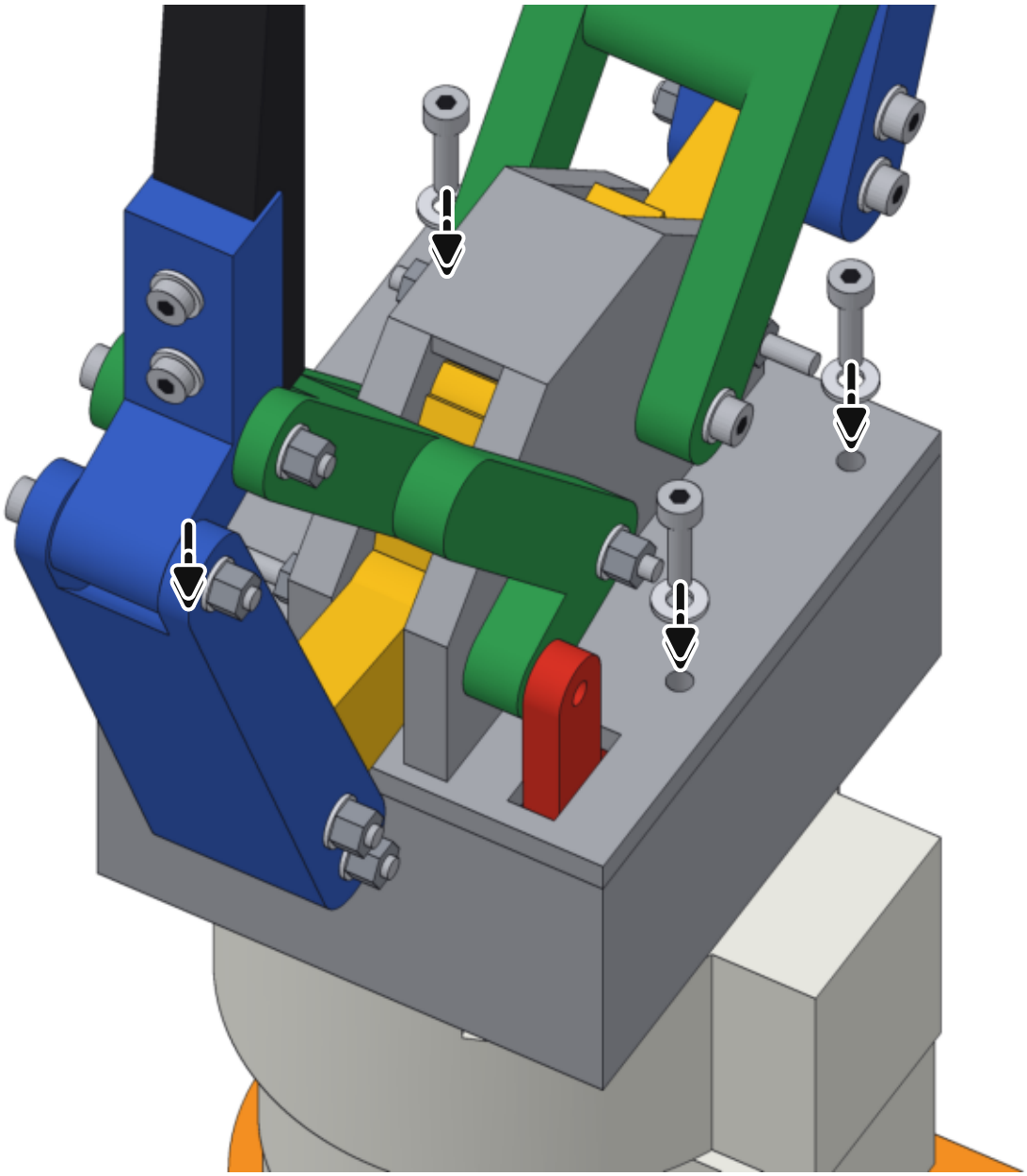
Tornillos de la torre



4x
M3 x 12

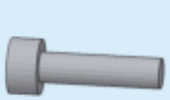


4x
Arandela M3

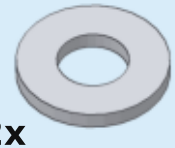


26

Biela a la oreja



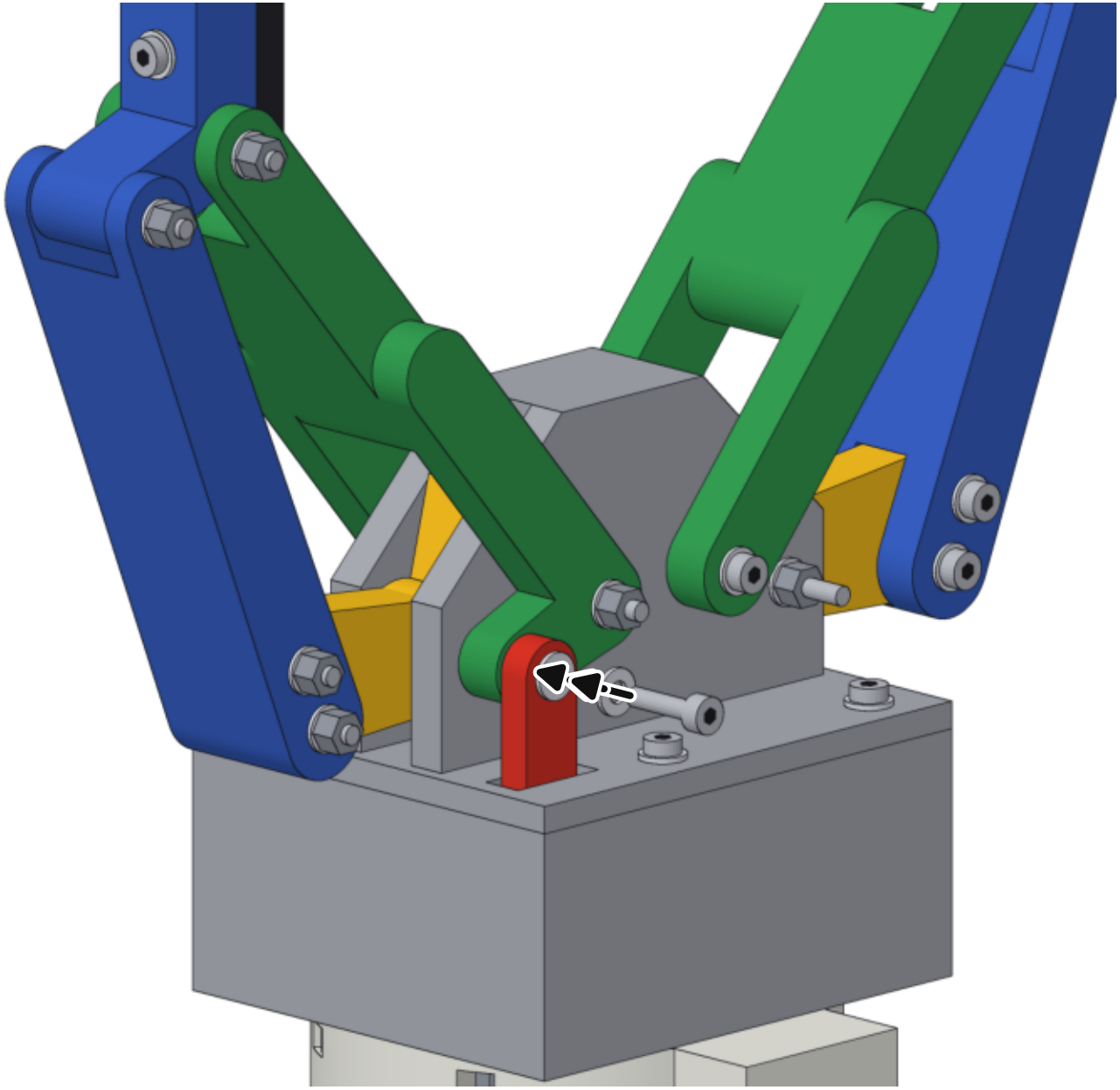
1x
M3 x 12



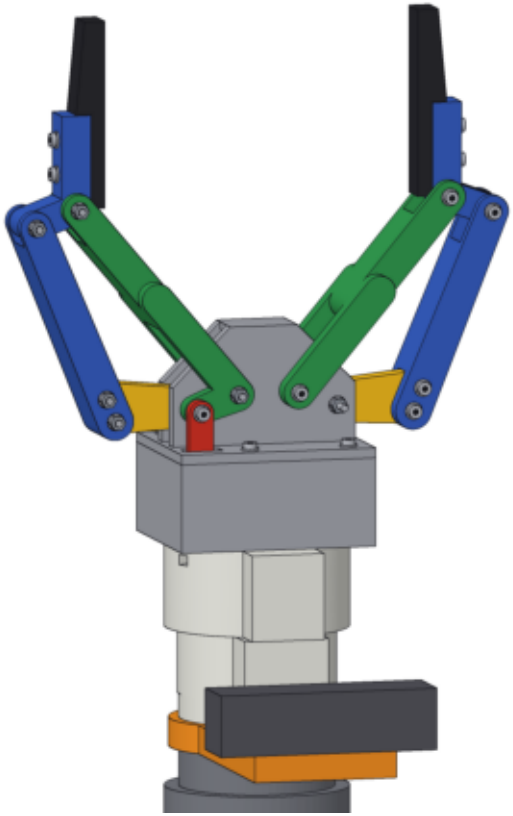
2x
Arandela M3

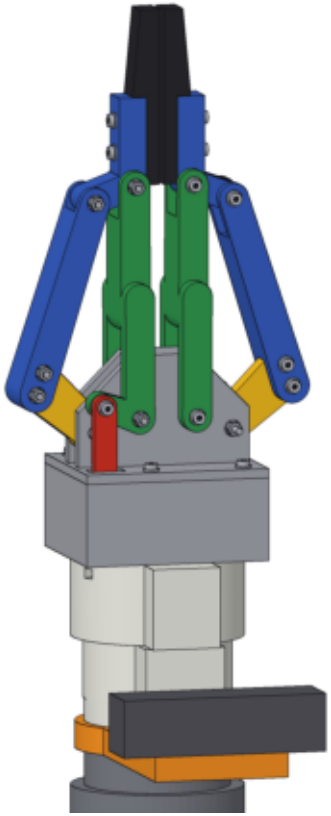
- Mueve los dedos a mano hasta alinear el agujero de la biela con la tuerca de la oreja.

- Apretar solo hasta quitar el juego.



Listo!





Primera prueba:

- Alimenta el servo a 6 V con limite de corriente
- Recorrido por software: 0 a +45,4 deg desde la posicion de montaje (q = 0 ... 0,7 del simulado
- Relacion casi lineal: servo_deg = 64,8 x q.
- Topes mecanicos en -2,0 y +45,7 deg.

Si algo roza: revisa que los dos engranes esten simetricos (paso 13) y que todas las tuercas de pivote giren libres.

Abierto: servo 0 deg -> 130,7 mm

Cerrado: servo +45,4 deg -> 2 mm